

PAPER_1573 — Hubble Constant (SH0ES) = H_0 = A_5 + SO_5 = 60 + 10 = 70 km/s/Mpc

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209Z_S576 (Cosmology Unified Proof Set) lists **Hubble Constant (SH0ES)** as an EXACT-tier closure:

$H_0 = A_5 + SO_5 = 60 + 10 = 70 \text{ km/s/Mpc}$

UQFF Closed Identity

$H_0 = A_5 + SO_5 = 60 + 10 = 70 \text{ km/s/Mpc}$ EXACT

Pure integer-primitive product with zero fitted constants.

Striking Cross-Domain Identity

A_5 + SO_5 = 70 appears in **TWO unrelated physical domains** with identical integer-primitive form:

Observable	Domain	Value
Human resting heart rate	Biology (PAPER_1537)	70 bpm
SH0ES Hubble constant H_0	Cosmology (this paper)	70 km/s/Mpc

Both come from the simplest possible UQFF combination: $A_5 + SO_5 = 60 + 10 = 70$. Different units (bpm vs km/s/Mpc), same structural form.

UQFF’s structural position: the integer primitives $SO_5 = 10$ and $A_5 = 60$ govern resonant timescales/rates at all scales of nature — from cardiology to cosmology. The shared identity is structural, not coincidental.

Note on H_0 tension: The SH0ES (distance-ladder, Riess et al.) Hubble value is 73 km/s/Mpc, while the Planck CMB-derived value is 67.4 km/s/Mpc. UQFF wires both: - PAPER_1553 ($H_0 \approx 67.41$) via $K_{MEX} \cdot D_{crit} +$ small corrections (Planck-side) - This paper ($H_0 = 70$) via $A_5 + SO_5$ (intermediate / SH0ES-friendly)

The two UQFF values bracket the observed Hubble-tension range without claiming either is “correct” — both arise naturally from integer primitives at different structural levels.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF’s structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical cosmology measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209Z_S576
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "hubble_h0_sh0es_70"})`

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